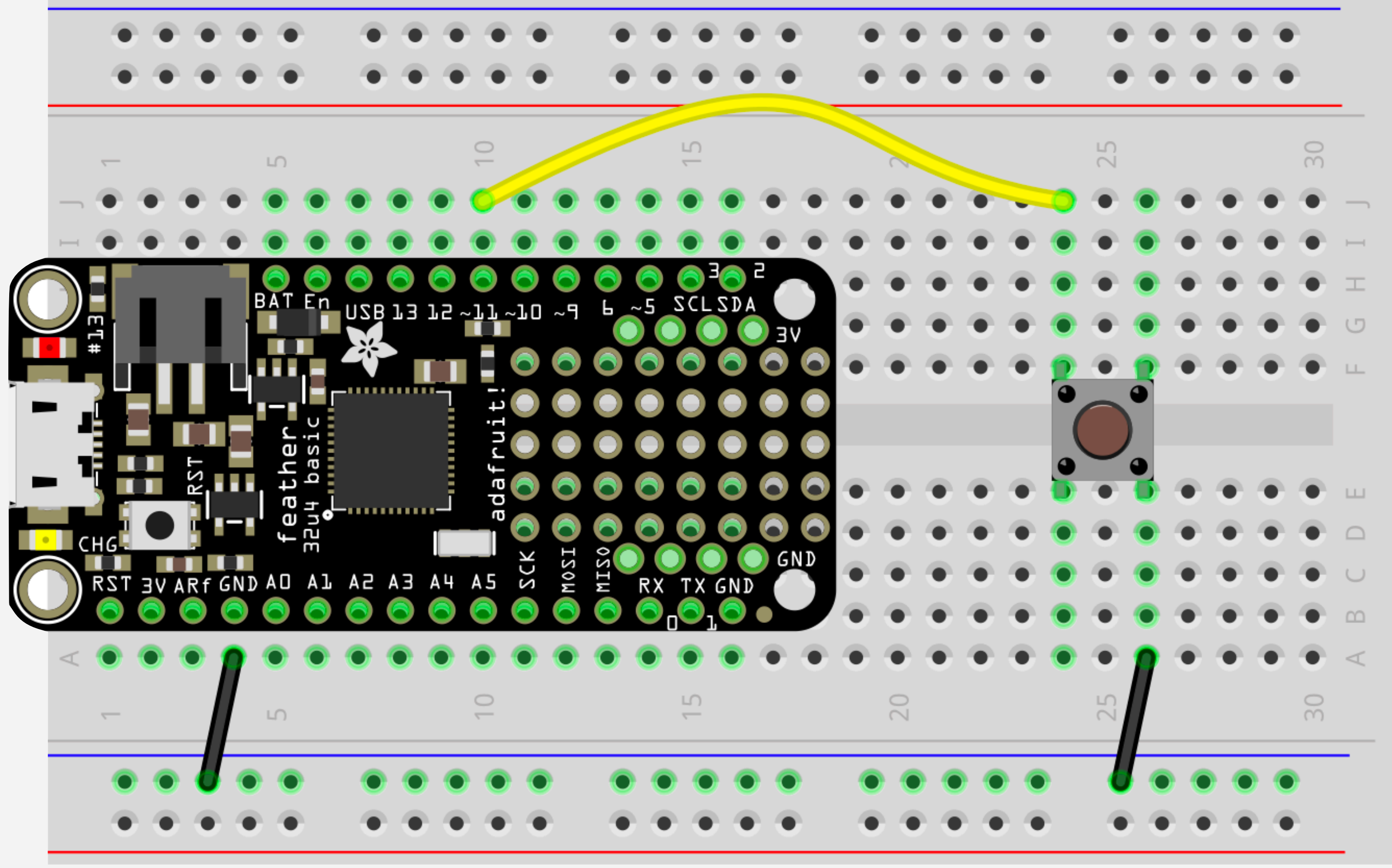


COMMUNICATING WITH THE COMPUTER

(HID) HUMAN INTERFACE DEVICE

Arudino -> Computer

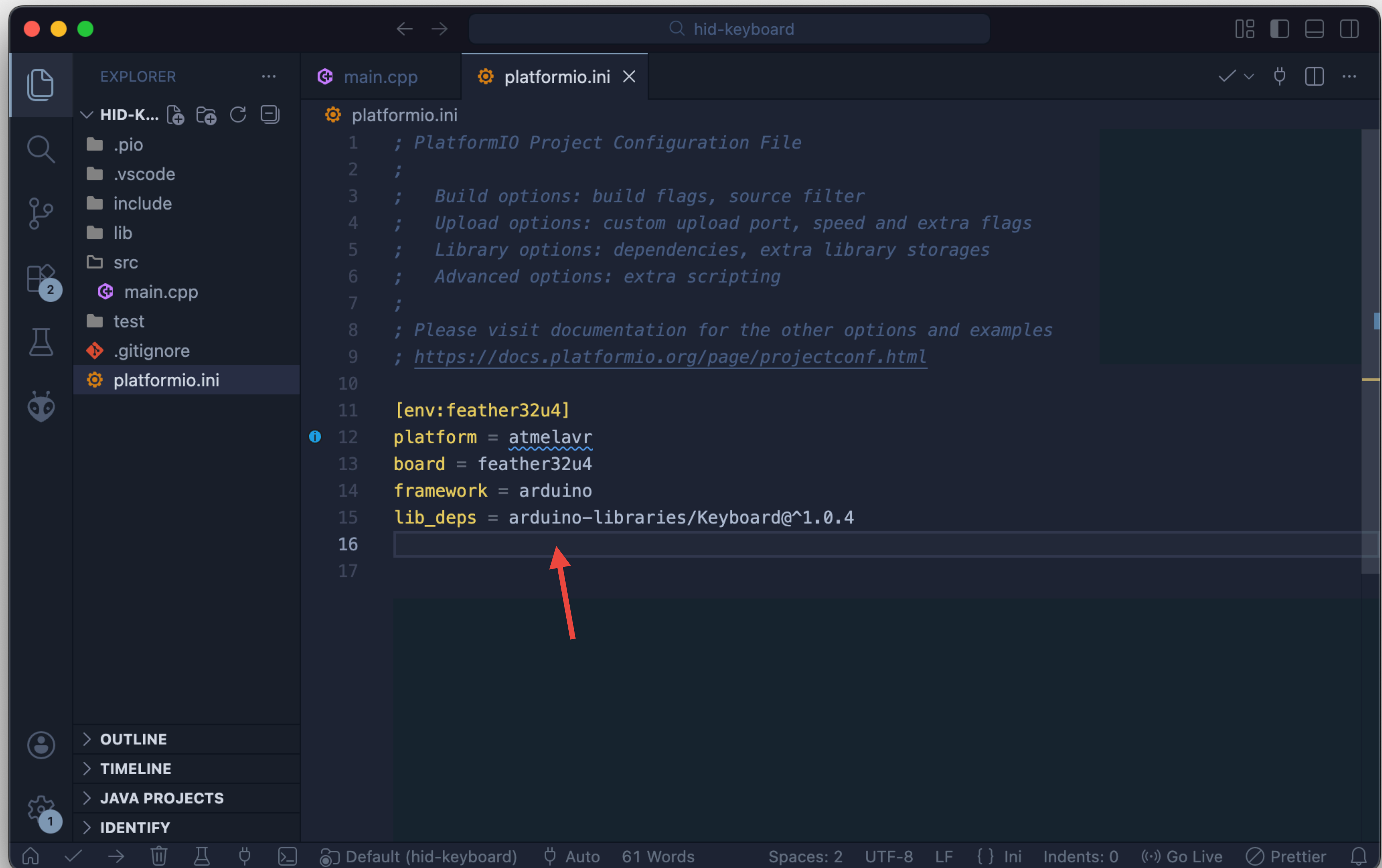


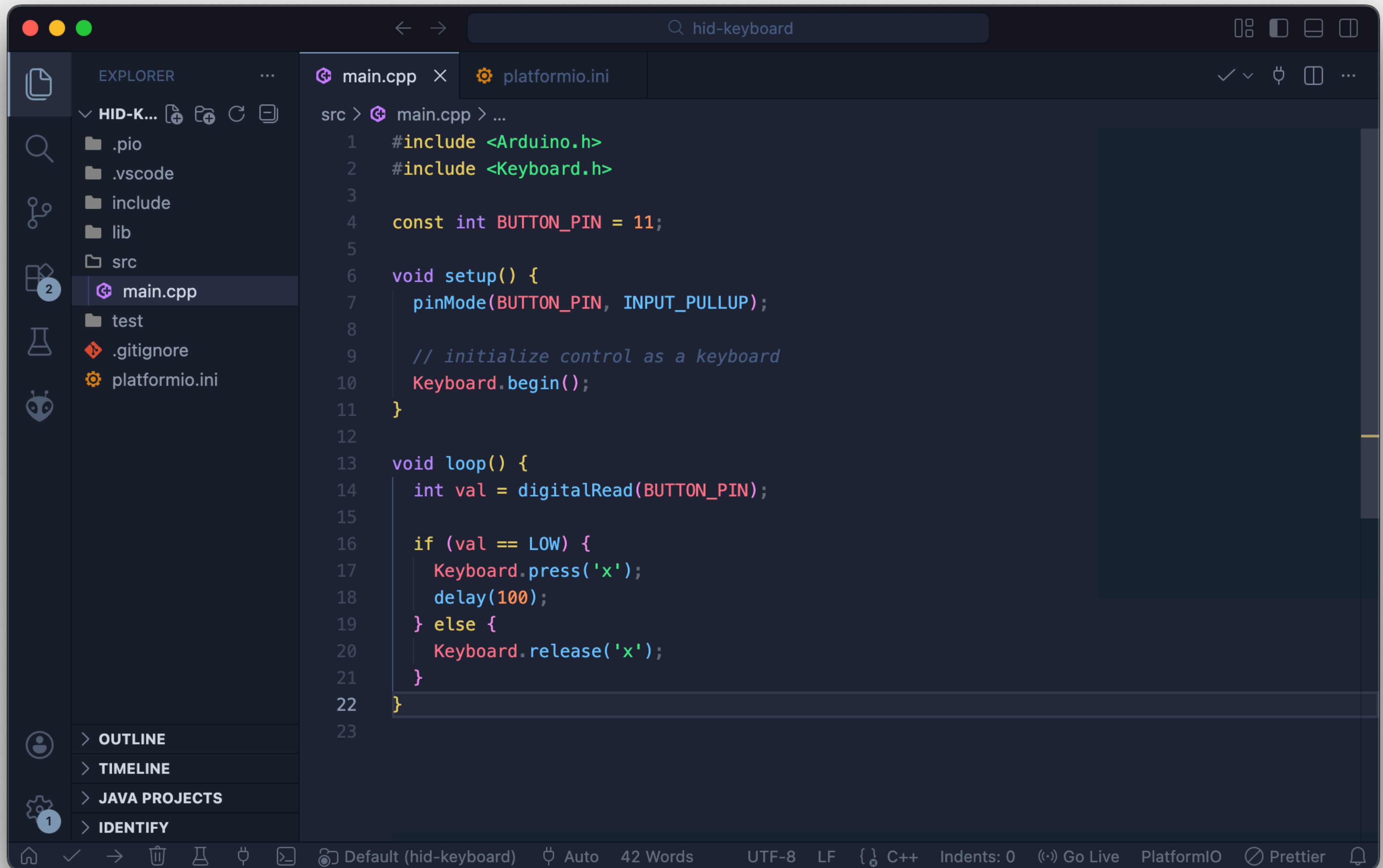
Using Libraries

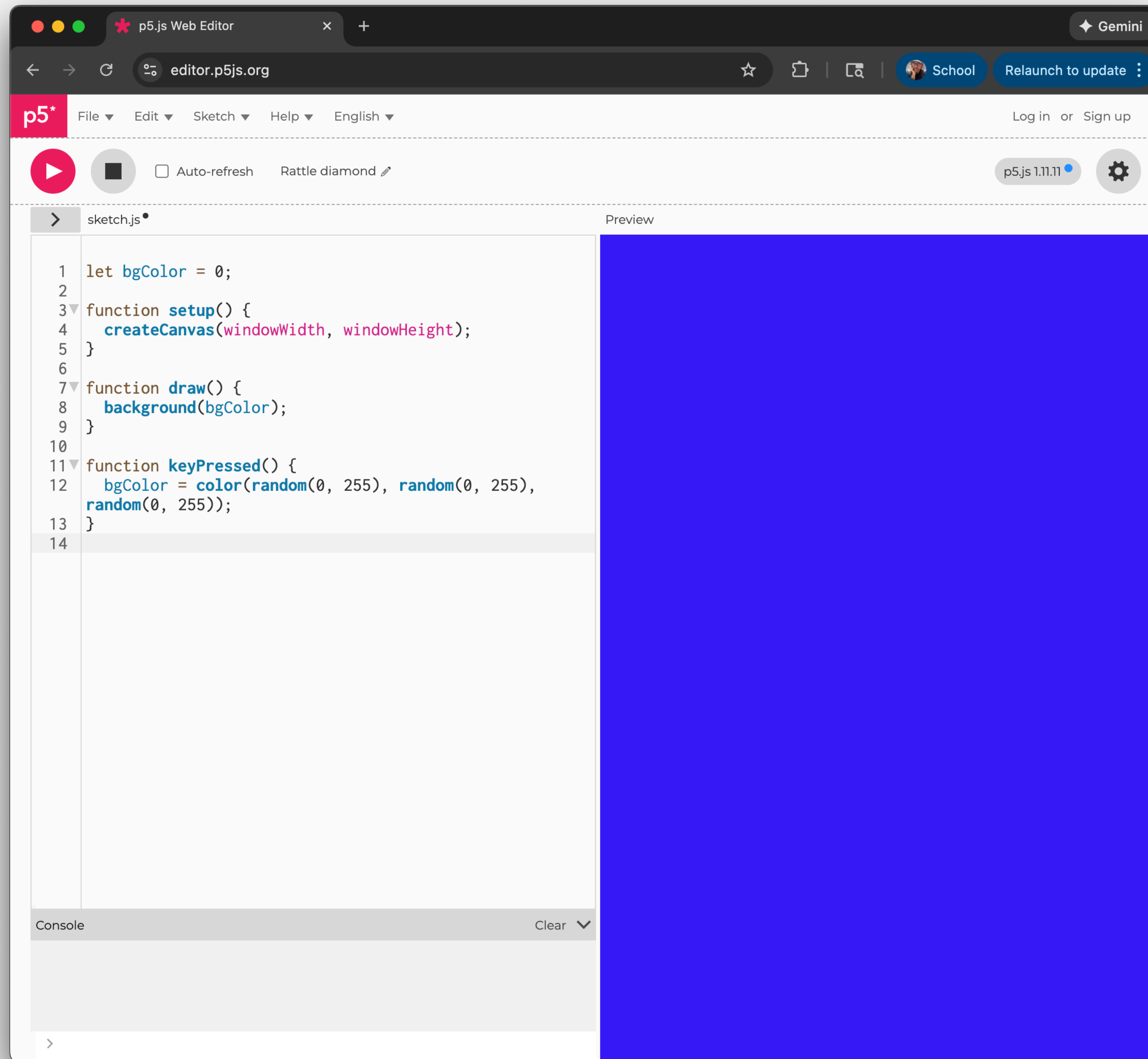
The core Arduino code can do lots of things! But not everything. To extend it's functionality it is occasionally necessary to use core or external libraries.

Using Libraries

Libraries are a way to extend functionality while keeping the code uncluttered. People can create libraries to do everything from wifi networking to debugging to support for unique devices







SERIAL

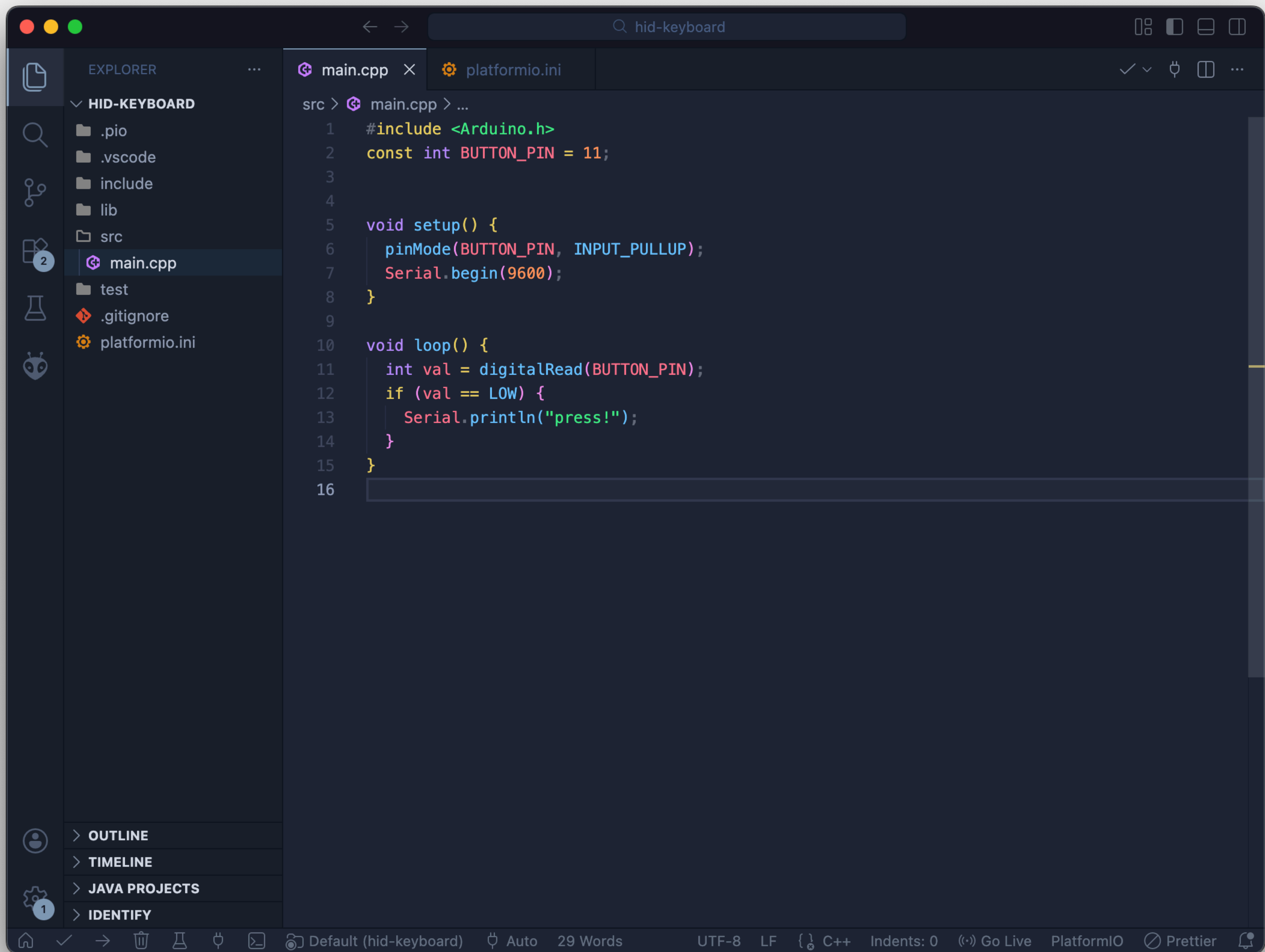
Arudino -> Computer

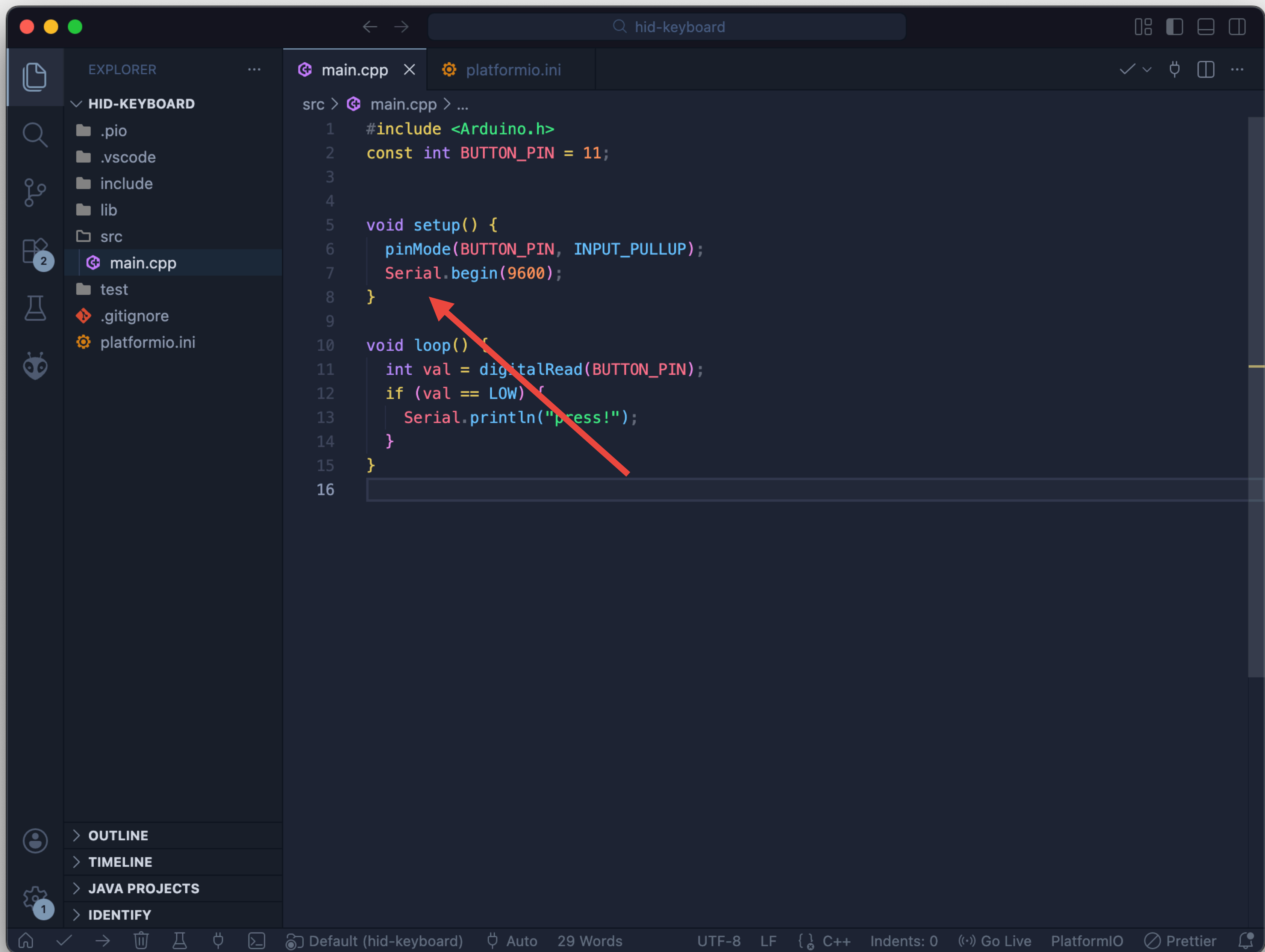
Debugging with Serial

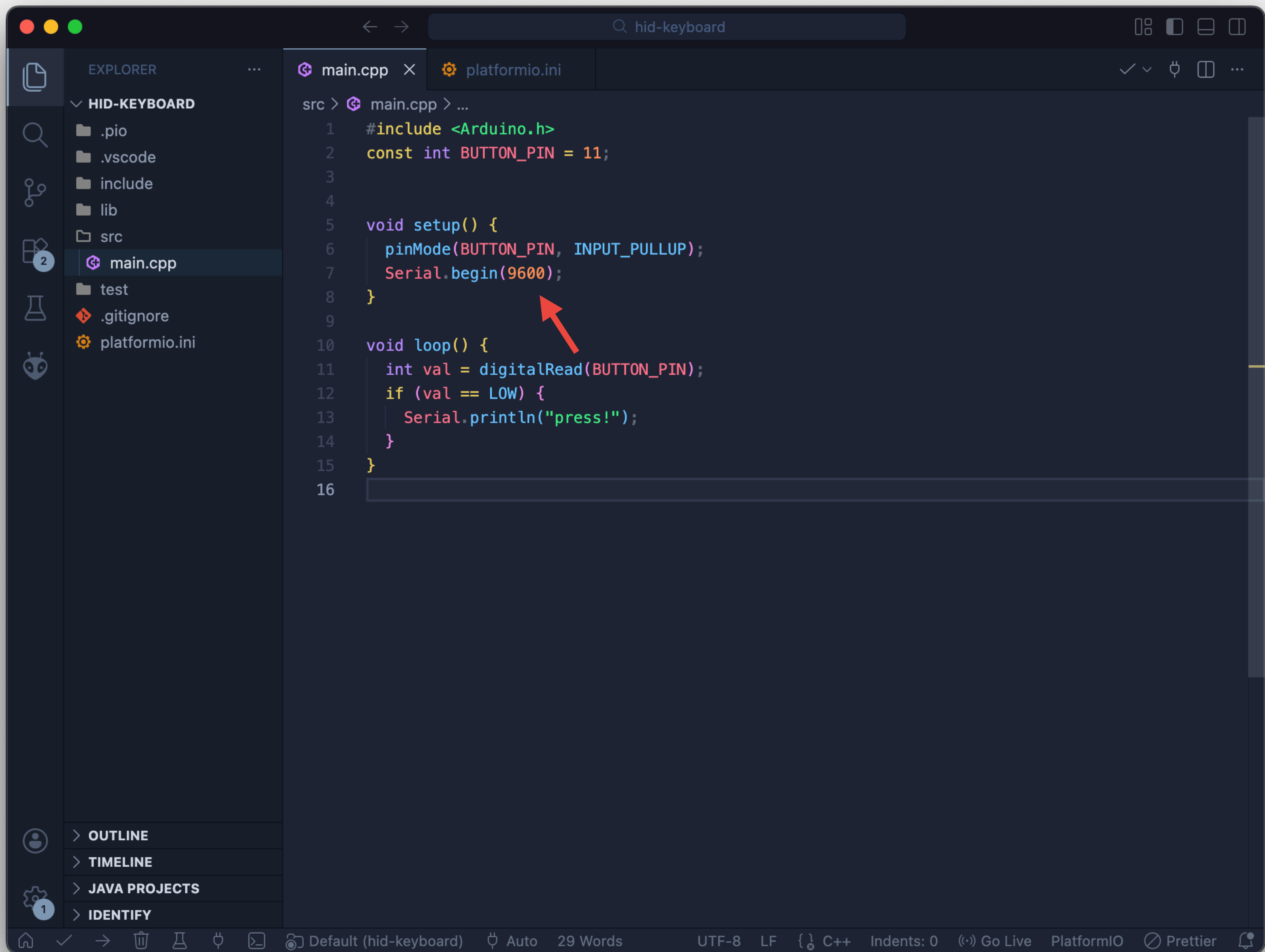
One of the most common ways
to use Serial is to help debug
your code.

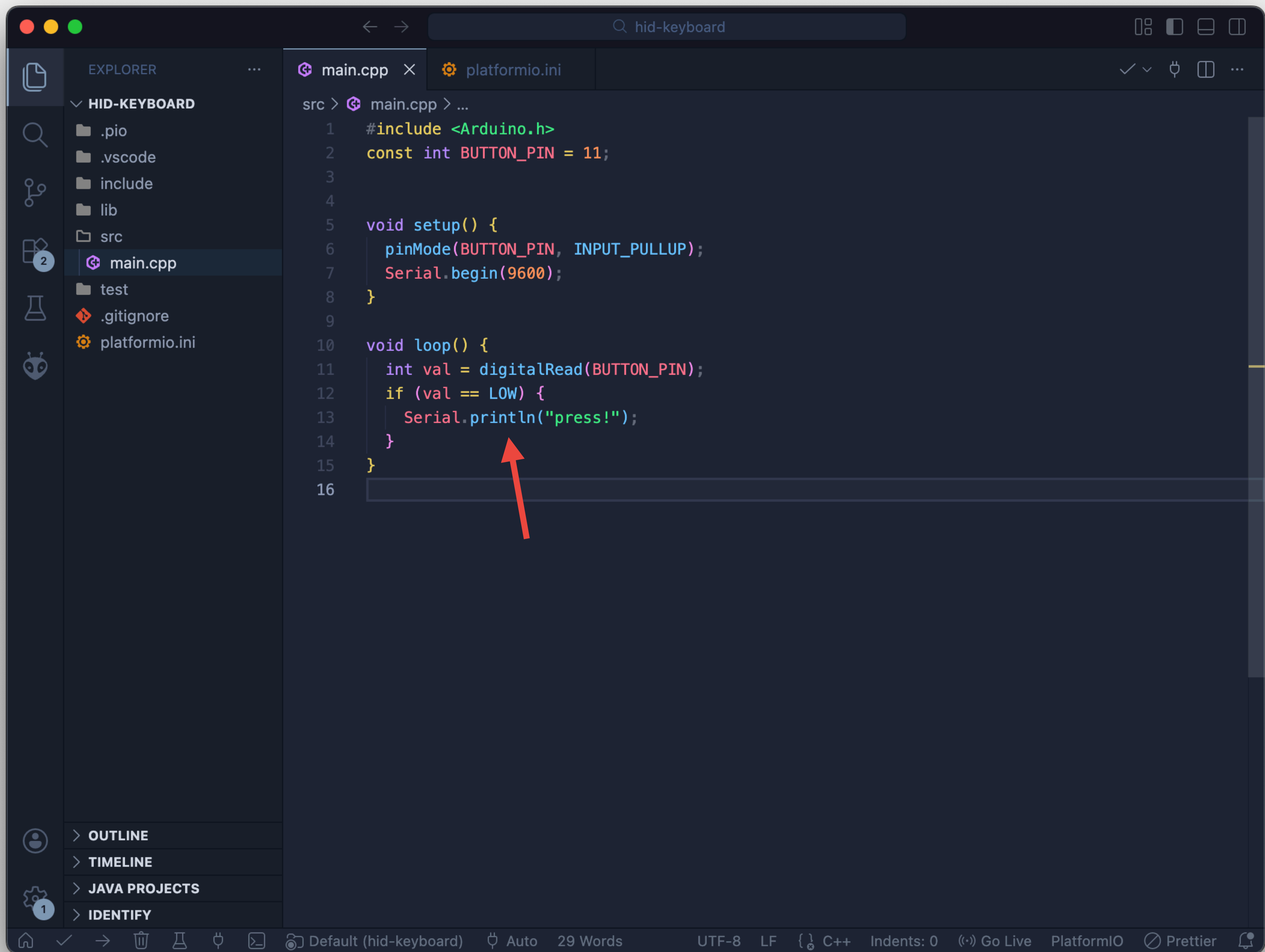
What is Serial?

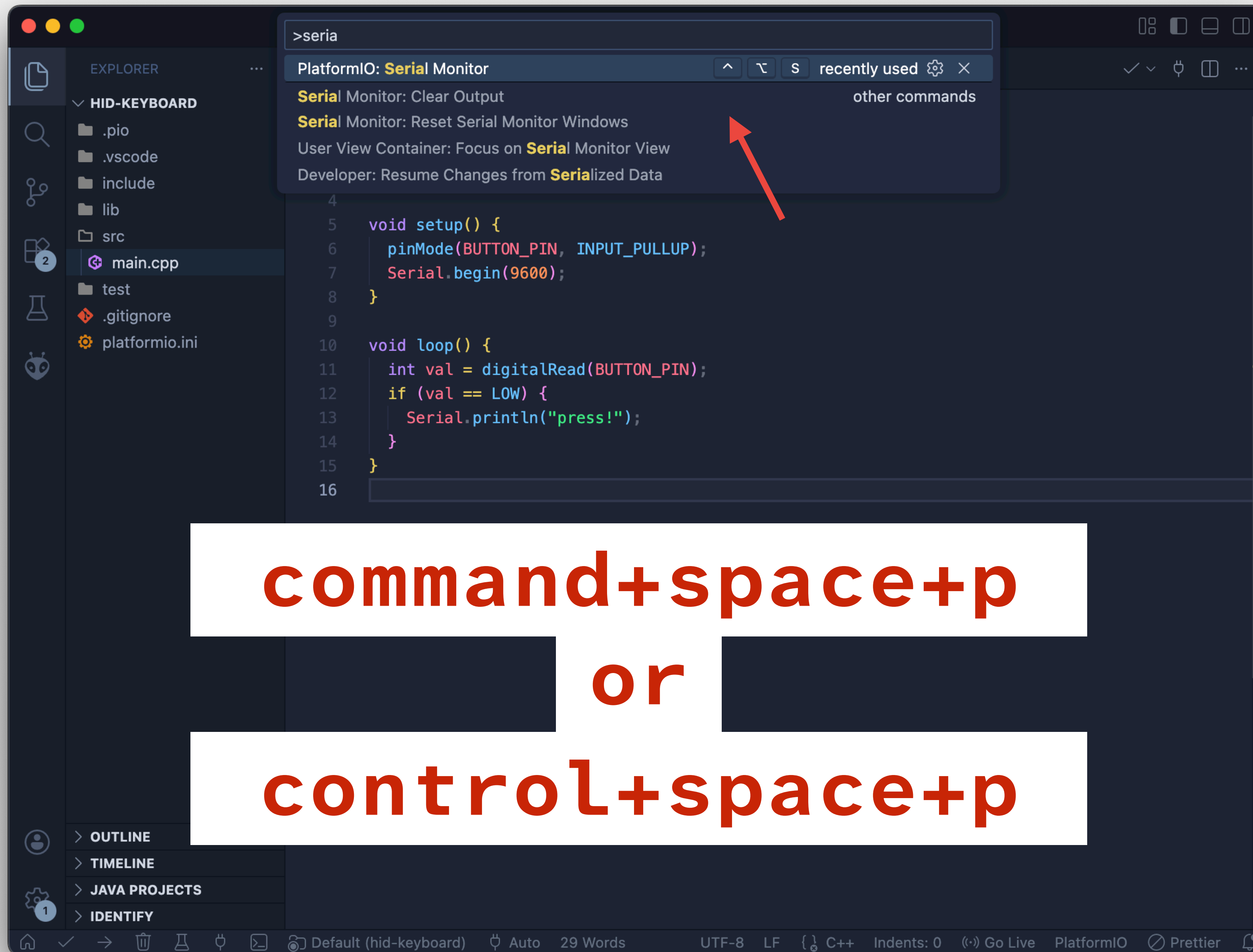
Serial communication is the process of sending data one bit at a time, sequentially, over a communication channel, in our case, the USB cord.







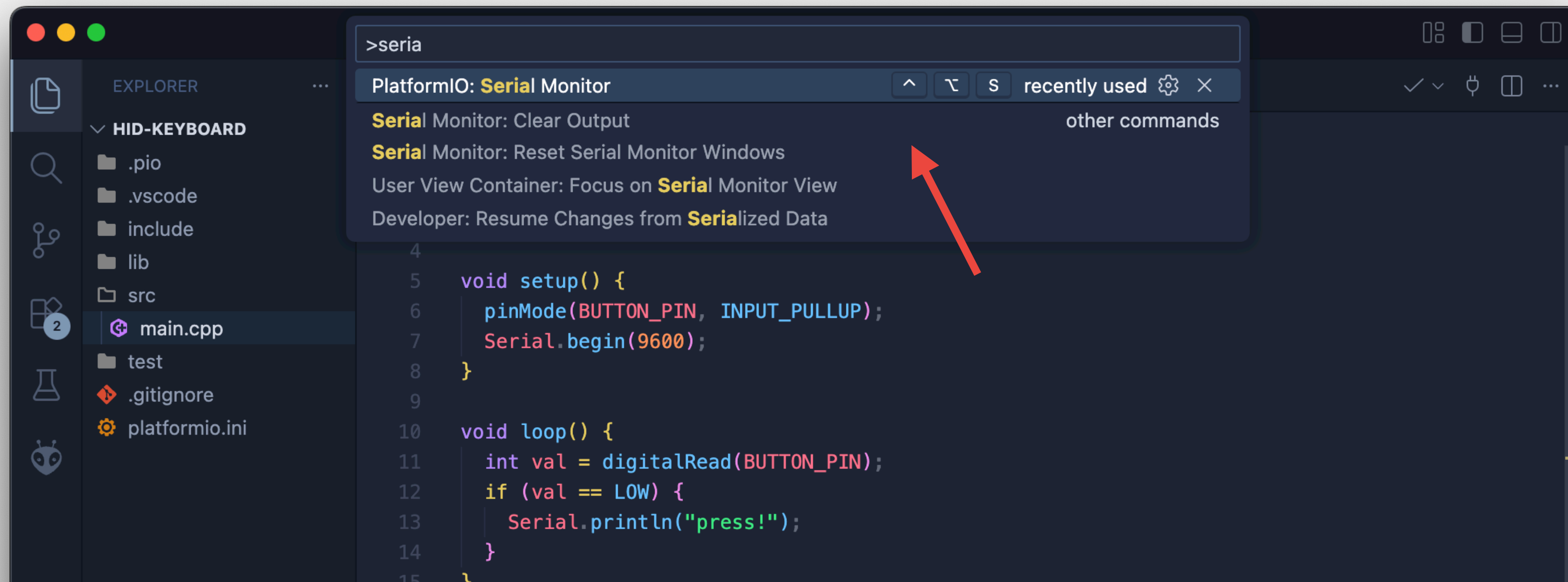




command+space+p

or

control+space+p



**Then type “serial” then
select
“PlatformIO: SerialMonitor”
and hit return**

EXPLORER

HID-KEYBOARD

.pio

.vscode

include

lib

src

main.cpp

test

.gitignore

platformio.ini

OUTLINE

TIMELINE

JAVA PROJECTS

IDENTIFY

main.cpp

platformio.ini

src > main.cpp > ...

1

#include <Arduino.h>

2

const int BUTTON_PIN = 11;

3

4

5

void setup() {

6

pinMode(BUTTON_PIN, INPUT_PULLUP);

7

Serial.begin(9600);

8

}

9

10

void loop() {

11

int val = digitalRead(BUTTON_PIN);

12

if (val == LOW) {

13

Serial.println("press!");

14

}

15

}

16

TERMINAL 7

TERMINAL

SERIAL MONITOR

+ Open an additional monitor

Monitor Mode Serial View Mode Text

Port /dev/tty.A30 - Baud rate 115200 Line ending None

Start Monitoring

Type in a message to send to the serial port. Send as Text Send Ctrl + C

PROBLEMS

OUTPUT

1

Default (hid-keyboard)

Auto

29 Words

Ln 16, Col 1

Spaces: 2

UTF-8

LF

C++

Indents: 0

Go Live

PlatformIO

Prettier

Who is Listening?

You can listen to Serial data
using all kinds of software,
Unity, p5 (on Chrome),
Processing, etc.